

Option market-making and vol arbitrage

Vladimir Lucic and Alex Tse introduce a market-making model for options that encompasses the trader's view on the underlying volatility versus the market-implied volatility surface. An approximately optimal strategy is derived in closed form, where the optimal bid and ask levels depend on the expected volatility arbitrage profits associated with the quoted options. The authors demonstrate how risk control over customised factors can be incorporated within the trading algorithm, and how doing so can improve the resilience of the market-making strategy against unexpected movements of the implied volatility surface

Options trading activity has experienced phenomenal growth over the last few years. In October 2023 Bloomberg reported that the global volume of options trading across multiple asset classes had surpassed that of futures (Doherty *et al* 2023). The rapid increase in options turnover presents both a challenge and an opportunity to market makers, whose operation requires split-second provision of competitive bid and ask quotes for a large number of option instruments (of different strikes and expiry dates on different underlyings) simultaneously. Efficient automation of option market-making is therefore of great practical interest.

Algorithmic market-making has received a lot of attention in the quantitative finance research community, but existing works mostly focus on 'delta one' instruments such as single stocks or futures. For a review of the standard mathematical frameworks for market-making, readers are referred to the excellent textbooks of Cartea *et al* (2015) and Guéant (2016). The question of how the existing approaches can be extended to option market-making, however, is far from trivial, because of the sheer size of the instruments universe and different risk exposures they create. Studies dedicated to option market-making are relatively scarce. Stoikov & Sağlam (2009) consider discrete-time approximation to compute optimal bid or ask quotes for a single European option. El Aoud & Abergel (2015) and Baldacci *et al* (2021) look at optimal market-making for single and multiple options, respectively, under a stochastic volatility model. Lai *et al* (2023) show that a Hawkes process can satisfactorily model option-market order flow, and they approximate the optimal quotes by a deep neural network.

In this paper, we present an option market-making model based on the model of Avellaneda & Stoikov (2008). A special feature of our framework is that it directly facilitates the expression of the market maker's view on volatility via the difference between their estimated actual volatility and the market-implied volatility, which is natural from the perspectives of derivatives trading and volatility arbitrage. In a baseline setup featuring only a single option, we derive closed-form expressions for the optimal bid and ask quotes under a second-order approximation. The optimal quotes admit a financially intuitive decomposition as a sum of three components:

- the expected volatility arbitrage profit that the market maker can harvest based on the actual-versus-implied volatility differential;
- a quantity linked to the demand-and-supply elasticity governing how sensitive the order flow is to a change in the quoted spreads; and
- a risk-adjustment term related to the penalty level imposed on the running and residual inventories.

We then discuss how the framework can be extended to market-making of multiple options by using a versatile factor-based approach to control the option portfolio risk. Finally, we demonstrate the practical implementation of the option market-making algorithm via a numerical example. Miscellaneous extensions and technical details omitted in this paper can be found in the extended preprint (Lucic & Tse 2024).

The baseline model

Given a market maker who provides pricing quotes for a European call option with a given strike $K > 0$ and maturity $\tau > 0$ written on a risky asset,¹ denote by $S = (S_t)_{t \geq 0}$ the price process of the underlying risky asset. From the perspective of the market maker, S has the dynamics $dS_t/S_t = \sigma_t dB_t$ where B is a standard one-dimensional Brownian motion. The market maker does not have any view on the (short-term) asset drift, while $\sigma = (\sigma_t)_{t \geq 0}$ is their subjective assessment of the asset volatility process. In practice, $(\sigma_t)_{t \geq 0}$ can originate from some volatility estimation or prediction procedure beyond the specification of our market-making model. We assume $\sigma_t = \sigma(t, S_t)$ for some function $\sigma(\cdot, \cdot)$ such that it has a local volatility form (and we shall see that a further specialisation to the constant volatility case can lead to a closed-form solution). Dividend and interest rates are assumed to be zero.

Option positions are evaluated on a mark-to-market basis according to a given implied volatility surface $\sigma_{\text{imp}} = \sigma_{\text{imp}}(K, \tau)$. For practical implementation, σ_{imp} can be inferred from either the market (mid-)prices of traded options or a suitably calibrated parametric model of the implied volatility surface. We assume σ_{imp} to be unchanged over the market-making horizon. This assumption is normally acceptable, as the implied volatility surface is typically recalibrated less frequently relative to the refresh of the limit order quotes. By the definition of implied volatility, the mark-to-market price of the call option at time t is given by:

$$O_t = O(t, S_t) := \text{BS}_{\text{call}}(S_t, K, \tau - t, \sigma_{\text{imp}}(K, \tau))$$

where BS_{call} is the Black-Scholes formula for call options. Alternatively, the function $O(t, s)$ can also be characterised by the solution to the Black-Scholes partial differential equation (PDE):

$$\partial_t O + \frac{1}{2} \sigma_{\text{imp}}^2 s^2 \partial_{ss} O = 0, \quad (t, s) \in [0, \tau) \times (0, \infty) \quad (1)$$

subject to the terminal condition $O(\tau, s) = (s - K)^+$.

Market-making takes place over a time interval $[0, T]$, with $T \in (0, \tau)$. At each time t , the market maker simultaneously posts buy and sell limit orders for one lot of the option at spreads δ_t^b and δ_t^a , respectively, around the mark-to-market option price; that is, the bid and ask prices are $O_t - \delta_t^b$ and $O_t + \delta_t^a$. Without loss of generality, the lot size is assumed to be 1. Let Y_t^b and Y_t^a represent the cumulative number of options that have been bought and sold, respectively, in the time interval $[0, t]$, and write the market maker's net option inventory at time t as Q_t , where we assume $Q_0 = 0$. Then by definition we have $Q_t = Y_t^b - Y_t^a$.

Following Avellaneda & Stoikov (2008), we model $Y^b = (Y_t^b)_{t \geq 0}$ and $Y^a = (Y_t^a)_{t \geq 0}$ as Cox processes with state-dependent intensities $\lambda_t^b = \lambda^b(\delta_t^b)$ and $\lambda_t^a = \lambda^a(\delta_t^a)$. We specify $\lambda^i(\delta^i) = \lambda_0^i e^{-\kappa^i \delta^i}$, where

¹ The analysis can be generalised to European puts or to any European derivative.

Order flow and inventory

Bid/ask quoted at spreads δ_t^b, δ_t^a around O_t . Fill intensities are Cox processes:

$$\lambda^i(\delta^i) = \lambda_0^i e^{-\kappa^i \delta^i}, \quad i \in \{b, a\}$$

Inventory: $Q_t = Y_t^b - Y_t^a$.

$\lambda_0^i, \kappa^i > 0$ are some constants for $i \in \{b, a\}$. Here, λ_0^i reflects the average number of arriving market orders per unit time if prices are quoted at the 'fair' mark-to-market level with zero spreads, while κ^i captures the elasticity of demand and supply such that the average number of arriving orders per unit time will drop by a fraction of $e^{-\kappa^i}$ when the quoted spread increases by one unit from zero.²

The market maker is required to delta hedge as part of their operational mandate, such that at time t they short Δ_t units of the underlying asset, with $\Delta_t := Q_t \partial_s O(t, S_t)$. The quantity $\partial_s O(t, S_t)$ is simply the Black-Scholes delta of the call option. Importantly, the computed delta is based on the implied volatility σ_{imp} , rather than the realised volatility.

The cash process of the market maker, denoted by $X = (X_t)_{t \geq 0}$, follows the dynamics:

$$X_t = \int_0^t (O_u + \delta_u^a) dY_u^a - \int_0^t (O_u - \delta_u^b) dY_u^b + \left[S_t \Delta_t - S_0 \Delta_0 - \int_0^t \Delta_u dS_u \right]$$

The first term represents the total sale proceeds when the ask prices are lifted, the second term the total purchase costs when the bid prices are hit, and the last term the cashflow from adjusting the delta hedge. The market maker's portfolio consists of cash, options and stocks, whose mark-to-market value at time t is given by $V_t := X_t + Q_t O_t - \Delta_t S_t$. The dynamics of V can be expressed as:

$$\begin{aligned} dV_t &= \delta_t^a dY_t^a + \delta_t^b dY_t^b + Q_t \partial_t O(t, S_t) dt \\ &\quad + \frac{1}{2} Q_t \sigma_t^2 S_t^2 \partial_{ss} O(t, S_t) dt \\ &= \frac{1}{2} (\sigma_t^2 - \sigma_{\text{imp}}^2) Q_t S_t^2 \partial_{ss} O(t, S_t) dt + \delta_t^a dY_t^a + \delta_t^b dY_t^b \end{aligned}$$

where we use the fact that, by construction, $\Delta_t := Q_t \partial_s O(t, S_t)$ and the Black-Scholes PDE (1), which allows us to write the option theta $\partial_t O$ in terms of its gamma $\partial_{ss} O$. The interpretation of the portfolio gain dV_t is financially natural: the first term is the volatility arbitrage profit depending on the realised-implied variance spread and the option gamma (also known as the gamma-theta carry), while the second and third terms are the spreads earned by the market maker when a bid or ask quote is filled.

The objective of the market maker is to maximise:

$$\mathbb{E} \left[V_T - \beta \int_0^T Q_u^2 du - \alpha Q_T^2 \right] \quad (2)$$

which is the expected value of the terminal portfolio value penalised by the terminal and running option inventories for some penalty parameters $\alpha \geq 0$ and $\beta \geq 0$. Practically, α controls the risk level at the end of a trading period, while β controls the intraday risk exposure throughout the entire duration of the market-making (and is linked to the risk-aversion level of an exponential utility function). Their values can be fine-tuned according to the underlying risk management policy. The market maker seeks a solution to the following dynamic optimisation problem:

$$h(t, s, q) := \sup_{\delta^b, \delta^a} \mathbb{E} \left[\int_t^T \left(\frac{\sigma_u^2 - \sigma_{\text{imp}}^2}{2} Q_u S_u^2 \partial_{ss} O(u, S_u) du \right. \right.$$

² See Fernandez-Tapia (2015) for a discussion of estimating λ_0^i and κ^i from market data.

$$\left. - \beta Q_u^2 du + \delta_u^a dY_u^a + \delta_u^b dY_u^b \right) - \alpha Q_T^2 \mid S_t = s, Q_{t-} = q \quad (3)$$

where the admissible controls $\delta^i = (\delta_t^i)_{t \geq 0}$ for $i \in \{b, a\}$ are some predictable and integrable processes such that $\mathbb{E}[\int_0^T |\delta_u^i| du] < \infty$.

■ **Solution to the option market-making problem.** Define the 'dollar gamma'³ of each option (evaluated at time t when the spot is s) as:

$$\Gamma^{\$}(t, s) := s^2 \partial_{ss} O(t, s) = \frac{K \phi(d_2(t, s))}{\sigma_{\text{imp}} \sqrt{\tau - t}}$$

where $\phi(\cdot)$ is the probability density function of a standard normal random variable and:

$$d_2(t, s) := \frac{\ln(s/K) - \sigma_{\text{imp}}^2(\tau - t)/2}{\sigma_{\text{imp}} \sqrt{\tau - t}}$$

The Hamilton-Jacobi-Bellman (HJB) equation associated with (3) is:

$$\begin{aligned} \partial_t h + \frac{1}{2} (\sigma^2(t, s) - \sigma_{\text{imp}}^2) \Gamma^{\$}(t, s) q - \beta q^2 + \frac{1}{2} \sigma^2(t, s) s^2 \partial_{ss} h \\ + \sup_{\delta^a} \{ \lambda^a(\delta^a) [\delta^a + h(t, s, q - 1) - h(t, s, q)] \} \\ + \sup_{\delta^b} \{ \lambda^b(\delta^b) [\delta^b + h(t, s, q + 1) - h(t, s, q)] \} = 0 \quad (4) \end{aligned}$$

with terminal condition $h(T, s, q) = -\alpha q^2$. Under our specification of the intensity functions $\lambda^i(\delta^i) = \lambda_0^i e^{-\kappa^i \delta^i}$ for $i \in \{b, a\}$, the optimal form of δ^b and δ^a can be pinned down from the first-order condition such that:

$$\left. \begin{aligned} \delta^{b,*} &= \frac{1}{\kappa^b} + h(t, s, q) - h(t, s, q + 1) \\ \delta^{a,*} &= \frac{1}{\kappa^a} + h(t, s, q) - h(t, s, q - 1) \end{aligned} \right\} \quad (5)$$

where the HJB equation (4) now becomes:

$$\begin{aligned} \partial_t h + \frac{\sigma^2(t, s) - \sigma_{\text{imp}}^2}{2} \Gamma^{\$}(t, s) q - \beta q^2 + \frac{\sigma^2(t, s)}{2} s^2 \partial_{ss} h \\ + \frac{\lambda_0^a e^{-1}}{\kappa^a} \exp[-\kappa^a (h(t, s, q) - h(t, s, q - 1))] \\ + \frac{\lambda_0^b e^{-1}}{\kappa^b} \exp[-\kappa^b (h(t, s, q) - h(t, s, q + 1))] = 0 \quad (6) \end{aligned}$$

Similarly to Avellaneda & Stoikov (2008), we consider a second-order expansion⁴ of the exponential terms such that (6) is approximated by:

$$\begin{aligned} \partial_t h + \frac{\sigma^2(t, s) - \sigma_{\text{imp}}^2}{2} \Gamma^{\$}(t, s) q - \beta q^2 + \frac{\sigma^2(t, s)}{2} s^2 \partial_{ss} h \\ + \frac{\lambda_0^a e^{-1}}{\kappa^a} \left[1 - \kappa^a (h(t, s, q) - h(t, s, q - 1)) \right. \\ \left. + \frac{(\kappa^a)^2}{2} (h(t, s, q) - h(t, s, q - 1))^2 \right] \\ + \frac{\lambda_0^b e^{-1}}{\kappa^b} \left[1 - \kappa^b (h(t, s, q) - h(t, s, q + 1)) \right. \\ \left. + \frac{(\kappa^b)^2}{2} (h(t, s, q) - h(t, s, q + 1))^2 \right] = 0 \end{aligned}$$

Quadratic ansatz and closed form

Guess $h(t, s, q) = \psi_0(t, s) + \psi_1(t, s)q + \psi_2(t, s)q^2$. Matching powers of q gives three coupled equations. The q^2 equation is a Riccati ODE:

$$\psi_2(t) = \sqrt{\Upsilon \beta} \frac{\zeta_- e^{-\eta(T-t)} - \zeta_+ e^{\eta(T-t)}}{\zeta_- e^{-\eta(T-t)} + \zeta_+ e^{\eta(T-t)}}$$

$$\Upsilon := \frac{1}{2e^{-1}(\lambda_0^a \kappa^a + \lambda_0^b \kappa^b)}, \quad \eta := \sqrt{\beta/\Upsilon}, \quad \zeta_{\pm} := \sqrt{\Upsilon \beta} \pm \alpha$$

The q^1 term $\psi_1(t, s)$ solves a linear parabolic PDE, solvable via Feynman-Kac:

$$\varphi(t, s) := \mathbb{E} \left[\int_t^T D(t, u) \frac{\sigma^2(u, S_u) - \sigma_{\text{imp}}^2}{2} \Gamma^{\$}(u, S_u) du \mid S_t = s \right]$$

which for constant real-world vol σ has closed form:

An ansatz for (7) is $h(t, s, q) = \psi_0(t, s) + \psi_1(t, s)q + \psi_2(t)q^2$ for some functions ψ_0 , ψ_1 and ψ_2 to be determined, whose **terminal conditions** are $\psi_0(T, s) = 0$, $\psi_1(T, s) = 0$ and $\psi_2(T) = -\alpha$. After substituting the ansatz into (7) and then equating the coefficients of the resulting polynomial in q to zero, we obtain:

$$\left. \begin{aligned} & \partial_t \psi_0 + \frac{\sigma^2(t, s)}{2} s^2 \partial_{ss} \psi_0 \\ & + \frac{\lambda_0^a e^{-1}}{\kappa^a} \left(1 - \kappa^a \psi_1 + \kappa^a \psi_2 + \frac{(\kappa^a)^2}{2} (\psi_1 - \psi_2)^2 \right) \\ & + \frac{\lambda_0^b e^{-1}}{\kappa^b} \left(1 + \kappa^b \psi_1 + \kappa^b \psi_2 + \frac{(\kappa^b)^2}{2} (\psi_1 + \psi_2)^2 \right) = 0 \\ & \partial_t \psi_1 + \frac{\sigma^2(t, s)}{2} s^2 \partial_{ss} \psi_1 + \frac{\sigma^2(t, s) - \sigma_{\text{imp}}^2}{2} \Gamma^S(t, s) \\ & + 2e^{-1} (\lambda_0^a \kappa^a + \lambda_0^b \kappa^b) \psi_2 \psi_1 + 2e^{-1} (\lambda_0^b - \lambda_0^a) \psi_2 \\ & + 2e^{-1} (\lambda_0^b \kappa^b - \lambda_0^a \kappa^a) \psi_2^2 = 0 \\ & \psi_2'(t) + \frac{\psi_2(t)^2}{\Upsilon} = \beta, \quad \psi_2(T) = -\alpha \quad \partial_t \psi_2 + 2e^{-1} (\lambda_0^b \kappa^b + \lambda_0^a \kappa^a) (\psi_2)^2 - \beta = 0 \end{aligned} \right\} \quad (8)$$

The last equation is a simple Riccati ordinary differential equation with terminal condition $\psi_2(T) = -\alpha$, whose solution is:

$$\psi_2(t) = \sqrt{\Upsilon \beta} \frac{\zeta_- e^{-\eta(T-t)} - \zeta_+ e^{\eta(T-t)}}{\zeta_- e^{-\eta(T-t)} + \zeta_+ e^{\eta(T-t)}} < 0 \text{ Always}$$

where:

$$\Upsilon := \frac{1}{2e^{-1} (\lambda_0^a \kappa^a + \lambda_0^b \kappa^b)}, \quad \eta := \sqrt{\frac{\beta}{\Upsilon}}, \quad \zeta_{\pm} := \sqrt{\Upsilon \beta} \pm \alpha$$

With the above expression for ψ_2 , the **second equation in (8) is now a linear parabolic PDE in the form:**

$$\partial_t \psi_1 + \frac{1}{2} \sigma^2(t, s) s^2 \partial_{ss} \psi_1 - r(t) \psi_1 + f(t, s) = 0$$

with:

$$\begin{aligned} r(t) &:= -2e^{-1} (\lambda_0^a \kappa^a + \lambda_0^b \kappa^b) \psi_2(t) \\ f(t, s) &:= \frac{\sigma^2(t, s) - \sigma_{\text{imp}}^2}{2} \Gamma^S(t, s) + 2e^{-1} (\lambda_0^b - \lambda_0^a) \psi_2(t) \\ &+ 2e^{-1} (\lambda_0^b \kappa^b - \lambda_0^a \kappa^a) \psi_2^2(t) \end{aligned}$$

Using the Feynman-Kac formula, its solution is found to be:

$$\begin{aligned} \psi_1(t, s) &= \varphi(t, s) + 2e^{-1} (\lambda_0^b - \lambda_0^a) \int_t^T D(t, u) \psi_2(u) du \\ &+ 2e^{-1} (\lambda_0^b \kappa^b - \lambda_0^a \kappa^a) \int_t^T D(t, u) \psi_2^2(u) du \end{aligned}$$

where:

$$\left. \begin{aligned} \varphi(t, s) &:= \mathbb{E} \left[\int_t^T D(t, u) \frac{\sigma^2(u, S_u) - \sigma_{\text{imp}}^2}{2} \Gamma^S(u, S_u) du \mid S_t = s \right] \\ D(t, u) &:= \frac{\zeta_- e^{-\eta(T-u)} + \zeta_+ e^{\eta(T-u)}}{\zeta_- e^{-\eta(T-t)} + \zeta_+ e^{\eta(T-t)}} \end{aligned} \right\} \quad (9)$$

In (9), if we ignore the factor $D(\cdot, \cdot)$ in the definition of φ , the latter is precisely the time- t expected total of the volatility arbitrage profit per unit long

position in the option with a delta hedge. In the important special case of the market maker's subjective volatility estimate **being a constant**, $\sigma_t \equiv \sigma$, we can compute $\mathbb{E}[\Gamma^S(u, S_u) \mid S_t = s]$ in closed form and in turn obtain:

$$\varphi(t, s) = \frac{K(\sigma^2 - \sigma_{\text{imp}}^2)}{2} \int_t^T \frac{D(t, u)}{\sqrt{\sigma_{\text{imp}}^2(\tau - u) + \sigma^2(u - t)}} \times \phi \left(\frac{\ln(s/K) - \frac{1}{2} \sigma^2(u - t) - \frac{1}{2} \sigma_{\text{imp}}^2(\tau - u)}{\sqrt{\sigma_{\text{imp}}^2(\tau - u) + \sigma^2(u - t)}} \right) du \quad (10)$$

See Ahmad & Wilmott (2005) for the derivation of the above expression.

Knowing the forms of ψ_1 and ψ_2 , we can deduce the **optimal spreads** from (5) as:

$$\begin{aligned} \text{bid} &= O_t - \delta^{b,*}, \quad \text{ask} = O_t + \delta^{a,*} \\ \delta^{b,*} &= \delta^{b,*}(t, s, q) = \frac{1}{\kappa^b} - \psi_1(t, s) - (2q + 1)\psi_2(t) \\ \delta^{a,*} &= \delta^{a,*}(t, s, q) = \frac{1}{\kappa^a} + \psi_1(t, s) + (2q - 1)\psi_2(t) \end{aligned} \quad (11)$$

The optimal quotes are completely analytical in the case of constant volatility such that the gamma-theta carry term $\varphi(t, s)$ is given by (10), but otherwise φ can still be conveniently computed by standard numerical methods.⁵ To gain insights into (11), we perform Taylor expansions in small $T - t$ (ie, a small market-making horizon, which is our main application focus) and write approximately $\alpha^2 \approx 0$. Informally, we have:

$$\begin{aligned} \delta^{b,*} &\approx \frac{1}{\kappa^b} - \varphi(t, s) \\ &+ \alpha(2q + 1 + 2e^{-1}(\lambda_0^b - \lambda_0^a)(T - t)) + \beta(2q + 1)(T - t) \\ \delta^{a,*} &\approx \frac{1}{\kappa^a} + \varphi(t, s) \\ &- \alpha(2q - 1 + 2e^{-1}(\lambda_0^b - \lambda_0^a)(T - t)) - \beta(2q - 1)(T - t) \end{aligned}$$

Thus, the financial interpretation of the optimal spreads can be roughly understood as follows. The first term, $1/\kappa^i$, is related to the elasticity of the options' demand and supply. The more sensitive the rate of the order flow is to the quoted spreads (ie, the larger the κ^i), the more competitive the provided quotes will be. The second component, φ , represents the 'edge' of the market maker in terms of the expected volatility arbitrage profit they can generate: if, for example, the trader expects to make money from volatility arbitrage via longing options because they believe the actual volatility will be higher than the implied volatility (ie, φ is large and positive), then they are happy with paying a higher price to acquire that position, and hence a more competitive (conservative) bid (ask) is quoted. Finally, the last two terms are some adjustment components governed by the penalty parameters α, β , the current inventory level q , the order flow imbalance $\lambda_0^b - \lambda_0^a$ and the remaining duration of the market-making horizon $T - t$.

Market-making of multiple options on the same underlying

Instead of trading a single call option, we now assume the market maker provides bid and ask quotes on N different call options written on the

⁵ We emphasise that the expressions in (11) are not theoretically optimal, due to the second-order approximation used. In the extended version of this paper (Lucic & Tse 2024), we compare the approximated optimal quotes against the 'ground truth' obtained from the numerical solutions of the original PDE in (6), and we observe very satisfactory performance over a short market-making horizon.

same underlying asset. For $n \in \{1, \dots, N\}$, suppose the n th option has strike K_n and maturity τ_n , such that its mark-to-market value at time t is $O_t^n = O^n(t, S_t)$ with $O^n(t, s) := \text{BS}_{\text{call}}(s, K_n, \tau_n - t, \sigma_{\text{imp}}(K_n, \tau_n))$. Again, we assume the market-making horizon T is short enough that $T < \min(\tau_1, \dots, \tau_N)$. The bid and ask prices quoted for the n th option are $O_t^n - \delta_t^{b,n}$ and $O_t^n + \delta_t^{a,n}$, respectively, where $(\delta_t^{a,n}, \delta_t^{b,n})_{t \geq 0}$ are the bid-ask spread processes for each option chosen by the market maker.

Let $Y_t^{b,n}$ and $Y_t^{a,n}$ be the cumulative units purchased and sold, respectively, for the n th option by time t . For each $n \in \{1, \dots, N\}$ and $i \in \{b, a\}$, we model $(Y_t^{i,n})_{t \geq 0}$ as Cox processes with intensity function $\lambda_t^{i,n} = \lambda_0^{i,n} \exp(-\kappa^{i,n} \delta_t^{i,n})$. The net inventory of the n th option at time t is $Q_t^n := Y_t^{b,n} - Y_t^{a,n}$. We write the inventory vector at time t as $\mathbf{Q}_t := [Q_t^1, \dots, Q_t^N]^T$.

As before, we require the market maker to delta hedge such that at time t they are required to short $\Delta_t := \sum_{n=1}^N Q_t^n \partial_s O^n(t, S_t)$ units of the underlying asset. Following the same analysis as in the previous section, the market maker's portfolio value process is:

$$dV_t = \sum_{n=1}^N \left[\frac{\sigma^2(t, s) - \sigma_{\text{imp}}^2(K_n, \tau_n)}{2} Q_t^n \Gamma^{s,n}(t, S_t) dt + \delta_t^{a,n} dY_t^{a,n} + \delta_t^{b,n} dY_t^{b,n} \right]$$

where $\Gamma^{s,n}$ is the dollar gamma of the n th option defined via $\Gamma^{s,n}(t, s) := s^2 \partial_{ss} O^n(t, s)$.

With multiple options, consider that the market maker seeks to maximise profit while minimising exposure to E number of risk factors, where each of them can be expressed as some linear combination of the risk weights of the individual options. Then the objective function can be formulated as:

$$\mathbb{E} \left\{ V_T - \int_0^T \sum_{k=1}^E \left[\beta_k \left(\sum_{n=1}^N \gamma_n^{(k)} \omega_n Q_u^n \right)^2 \right] du - \sum_{k=1}^E \left[\alpha_k \left(\sum_{n=1}^N \gamma_n^{(k)} \omega_n Q_T^n \right)^2 \right] \right\} \quad (12)$$

In the above, $\{\gamma_n^{(k)}\}_{n=1, \dots, N; k=1, \dots, E}$ define the construction of each risk factor, $\{\omega_n\}_{n=1, \dots, N}$ are the risk weights of each option, and $\alpha_k \geq 0$ and $\beta_k \geq 0$, respectively, are the terminal and running-penalty parameters associated with the k th risk factor. This specification is simple yet flexible enough to incorporate many practical risk management considerations. For example, we might wish to protect the option portfolio from structural changes in the implied volatility surface, such as parallel shift and movement in skew. Exposure to parallel shift can be captured by the total vega across all instruments: we choose the risk factor to be $\sum_{n=1}^N \text{Vega}_0^n Q_T^n$, where Vega_0^n is the initial vega of option n . Meanwhile, exposure to skew can be proxied by the vega differential of options at the two extremes of moneyness. An example of an appropriate risk factor could be:

$$\sum_{n=1}^N (1_{\{K_n \leq 0.9 S_0\}} - 1_{\{K_n \geq 1.1 S_0\}}) \text{Vega}_0^n Q_T^n$$

where K_n is the strike of option n .⁶ If we want to simultaneously penalise exposure to parallel shift as well as changes in the skewness of the implied

volatility surface, then in (12) we can set $E = 2$, $\omega_n = \text{Vega}_0^n$, $\gamma_n^{(1)} = 1$ for all n and $\gamma_n^{(2)} = 1_{\{K_n \leq 0.9 S_0\}} - 1_{\{K_n \geq 1.1 S_0\}}$. In the 'Numerical example' subsection, we shall give another example involving risk-partitioning of multiple options based on their maturities and moneynesses.

■ **Solution to the multi-option market-making problem.** Let $\mathbf{R}_\alpha \in \mathbb{R}^{E \times N}$ and $\mathbf{R}_\beta \in \mathbb{R}^{E \times N}$ be matrixes whose (k, n) elements are given, respectively, by:

$$[\mathbf{R}_\alpha]_{k,n} := \gamma_n^{(k)} \omega_n \sqrt{\alpha_k} \quad \text{and} \quad [\mathbf{R}_\beta]_{k,n} := \gamma_n^{(k)} \omega_n \sqrt{\beta_k}$$

Define the symmetric positive definite matrixes $\mathbf{A} := \mathbf{R}_\alpha^T \mathbf{R}_\alpha \in \mathbb{R}^{N \times N}$ and $\mathbf{B} := \mathbf{R}_\beta^T \mathbf{R}_\beta \in \mathbb{R}^{N \times N}$. The value function can then be written as:

$$h(t, s, \mathbf{q}) := \sup_{\delta^{i,n}} \mathbb{E} \left[\int_t^T \left(\sum_{n=1}^N \left[\frac{\sigma^2(t, s) - \sigma_{\text{imp}}^2(K_n, \tau_n)}{2} Q_u^n \Gamma^{s,n}(u, S_u) du + \delta_u^{a,n} dY_u^{a,n} + \delta_u^{b,n} dY_u^{b,n} \right] - \mathbf{Q}_u^T \mathbf{B} \mathbf{Q}_u du \right) - \mathbf{Q}_T^T \mathbf{A} \mathbf{Q}_T \mid S_t = s, \mathbf{Q}_{t-} = \mathbf{q} \right] \quad (13)$$

Its associated HJB equation is:

$$\begin{aligned} \partial_t h + \frac{\sigma^2(t, s) s^2}{2} \partial_{ss} h + \mathbf{C}^T(t, s) \mathbf{q} - \mathbf{q}^T \mathbf{B} \mathbf{q} \\ + \sum_{n=1}^N \left\{ \sup_{\delta^{a,n}} \lambda^{a,n}(\delta^{a,n}) [\delta^{a,n} + h(t, s, \mathbf{q} - \mathbf{e}_n) - h(t, s, \mathbf{q})] \right\} \\ + \sum_{n=1}^N \left\{ \sup_{\delta^{b,n}} \lambda^{b,n}(\delta^{b,n}) [\delta^{b,n} + h(t, s, \mathbf{q} + \mathbf{e}_n) - h(t, s, \mathbf{q})] \right\} = 0, \end{aligned}$$

subject to terminal condition $h(T, s, \mathbf{q}) = -\mathbf{q}^T \mathbf{A} \mathbf{q}$. In the above, $\mathbf{C}(t, s) \in \mathbb{R}^N$ is a vector-valued function where:

$$[\mathbf{C}(t, s)]_n := \frac{\sigma^2(t, s) - \sigma_{\text{imp}}^2(K_n, \tau_n)}{2} \Gamma^{s,n}(t, s)$$

is the volatility arbitrage profit rate per unit option n , and each $\mathbf{e}_n \in \mathbb{R}^N$ is a basis vector (ie, a vector with a value of unity for its n th entry and zero elsewhere). Under $\lambda^{i,n}(\delta) = \lambda_0^{i,n} e^{-\kappa^{i,n} \delta}$, the optimal controls are:

$$\begin{aligned} \delta^{a,n,*} &= \frac{1}{\kappa^{a,n}} + h(t, s, \mathbf{q}) - h(t, s, \mathbf{q} - \mathbf{e}_n) \\ \delta^{b,n,*} &= \frac{1}{\kappa^{b,n}} + h(t, s, \mathbf{q}) - h(t, s, \mathbf{q} + \mathbf{e}_n) \end{aligned} \quad (14)$$

The extended version of this paper (Lucic & Tse 2024) gives further technical details about how to solve the problem. Doing so involves a second-order approximation similar to (7), followed by an ansatz in the form:

$$h(t, s, \mathbf{q}) = \theta_0(t, s) + \boldsymbol{\theta}_1^T(t, s) \mathbf{q} + \mathbf{q}^T \boldsymbol{\Theta}_2(t) \mathbf{q}$$

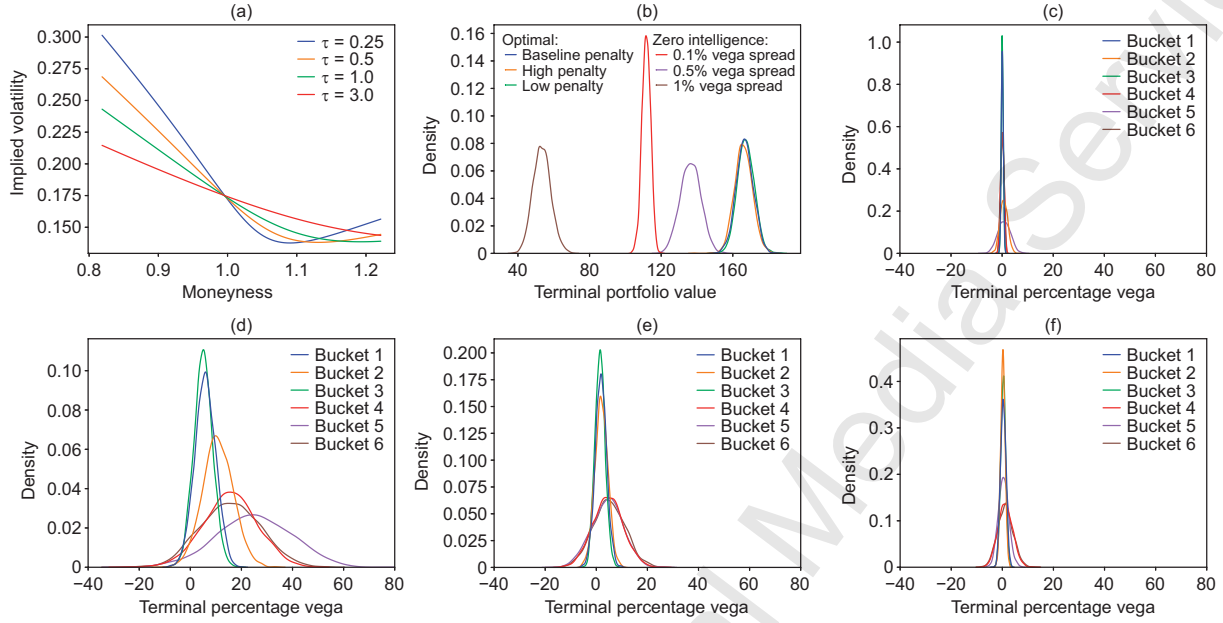
where $\theta_0(t, s) \in \mathbb{R}$, $\boldsymbol{\theta}_1(t, s) \in \mathbb{R}^N$ and $\boldsymbol{\Theta}_2(t) \in \mathbb{R}^{N \times N}$ are some (scalar-, vector- and symmetric matrix-valued) functions to be determined, with terminal conditions $\theta_0(T, s) = 0$, $\boldsymbol{\theta}_1(T, s) = \mathbf{0}$ and $\boldsymbol{\Theta}_2(T) = -\mathbf{A}$. We can subsequently derive a system of PDEs satisfied by θ_0 , $\boldsymbol{\theta}_1$ and $\boldsymbol{\Theta}_2$.

The optimal spreads (in vector form) are given by:

$$\begin{aligned} \delta^{b,*}(t, s, \mathbf{q}) &= \frac{1}{\kappa^b} - \boldsymbol{\theta}_1(t, s) - \text{diag}(\boldsymbol{\Theta}_2(t)) \mathbf{1} - 2\boldsymbol{\Theta}_2(t) \mathbf{q} \\ \delta^{a,*}(t, s, \mathbf{q}) &= \frac{1}{\kappa^a} + \boldsymbol{\theta}_1(t, s) - \text{diag}(\boldsymbol{\Theta}_2(t)) \mathbf{1} + 2\boldsymbol{\Theta}_2(t) \mathbf{q} \end{aligned} \quad (15)$$

⁶ We have implicitly assumed that each option's vega is approximately unchanged over the market-making horizon.

1 The implied volatility surface, and the empirical densities of the simulated terminal portfolio values and the vegas for each bucket



A. Options are partitioned into six groups (buckets) based on their maturities (long/short) and moneynesses (low/moderate/high)

Range of maturity τ	Range of moneyness $\tilde{K} := K/S_0$		
	(0, 0.95]	(0.95, 1.05]	[1.05, ∞)
(0, 9/12)	Bucket 1	Bucket 2	Bucket 3
[9/12, ∞)	Bucket 4	Bucket 5	Bucket 6

Note that the optimal market-making rules for option n generally depend on the state value of the entire inventory vector $\mathbf{q} = [q^1, q^2, \dots, q^N]^T$, rather than just the inventory level of that individual option q^n . This is because the portfolio effect is now taken into account based on the construction of the underlying risk factors. In the special case that $\mathbf{B} = \mathbf{0}_{N \times N}$, which arises, for example, when there is no running penalty (ie, $\beta_k = 0$ for all $k \in \{1, \dots, E\}$), we can indeed deduce:

$$\boldsymbol{\Theta}_2(t) = -\mathbf{A}(\mathbf{I} + 2e^{-1}(T-t)\text{diag}(\lambda_0^b \odot \kappa^b + \lambda_0^a \odot \kappa^a)\mathbf{A})^{-1} \quad (16)$$

where $\mathbf{I}$ is an identity matrix; $\lambda_0^i := [\lambda_0^{i,1}, \lambda_0^{i,2}, \dots, \lambda_0^{i,N}]^T$, $\kappa^i := [\kappa^{i,1}, \kappa^{i,2}, \dots, \kappa^{i,N}]^T$ for $i \in \{b, a\}$; $\odot$ denotes the Hadamard product of two vectors; $\mathbf{1}$ denotes a column vector with unit entries; and $\text{diag}(\mathbf{M})$ represents an N -by- N diagonal matrix whose diagonal entries are either $(M_1, \dots, M_N)$ if $\mathbf{M} \in \mathbb{R}^N$ or $(M_{11}, \dots, M_{NN})$ if $\mathbf{M} \in \mathbb{R}^{N \times N}$. We can further introduce the approximation that the dollar gamma is roughly unchanged over the market-making horizon, such that:

$$\begin{aligned} [C(t, s)]_n &= \frac{\sigma^2 - \sigma_{\text{imp}}^2(K_n, \tau_n)}{2} \Gamma^{\$, n}(t, s) \\ &\approx \frac{\sigma^2 - \sigma_{\text{imp}}^2(K_n, \tau_n)}{2} \Gamma^{\$, n}(0, S_0) =: [C_0]_n \end{aligned}$$

for all n, t and s . Here, $\mathbf{C}_0 \in \mathbb{R}^N$ is a constant vector collecting the estimated expected volatility arbitrage rate of each option computed by the initial gamma. Then, under this approximation, $\boldsymbol{\theta}_1(t, s)$ no longer depends on s and we have the closed-form solution:

$$\begin{aligned} \boldsymbol{\theta}_1(t, s) &= \boldsymbol{\theta}_1(t) = (\mathbf{I} + 2e^{-1}(T-t)\mathbf{A}\mathbf{D})^{-1} \\ &\quad \times \int_t^T (\mathbf{I} + 2e^{-1}(T-u)\mathbf{A}\mathbf{D}) \mathbf{f}(u) du \quad (17) \end{aligned}$$

where $\mathbf{f}(t) := 2e^{-1}\boldsymbol{\Theta}_2(t)(\lambda_0^b - \lambda_0^a) + 2e^{-1}\boldsymbol{\Theta}_2(t)\text{diag}(\boldsymbol{\Theta}_2(t))(\lambda_0^b \odot \kappa^b - \lambda_0^a \odot \kappa^a) + \mathbf{C}_0$ and $\mathbf{D} := \text{diag}(\lambda_0^b \odot \kappa^b + \lambda_0^a \odot \kappa^a)$.

Numerical example. As an illustration, we consider a market maker overseeing multiple vanilla options of different strikes and expiries on the same underlying. The available expiries are three months, six months, one year and three years (ie, $\tau \in \{3/12, 6/12, 1, 3\}$) and the available moneynesses range from 90% to 110% in increments of 2% (ie, $\tilde{K} := K/S_0 \in \{0.9, 0.92, 0.94, \dots, 1.08, 1.1\}$). Hence there are in total $N = 44$ different option instruments. In our setting, each option can be a call or a put.

Suppose the risk monitoring of the option portfolio is performed via bucketing. Specifically, options of similar strikes and maturities are grouped together and separate risk penalties will be applied to different buckets. We construct $E = 6$ buckets based on the rules shown in table A. The risk measure for each bucket is chosen to be the total percentage-vega of all instruments in the group. The penalty term for bucket k is chosen as $\alpha_k(\sum_{n \in \mathcal{B}_k} Q_T^n \times 1\% \times \text{Vega}_0^n)^2$, where $\mathcal{B}_k$ denotes the collection of options in bucket k and Vega_0^n is the initial vega of an individual option. As a baseline, we pick $\{\alpha_1, \alpha_2, \alpha_3, \alpha_4, \alpha_5, \alpha_6\} = \{0.2, 0.01, 0.2, 0.2, 0.01, 0.2\}$ such that buckets 1, 3, 4 and 6 receive a larger risk penalty so as to

discourage vega exposure at more extreme strikes. Meanwhile, all running-penalty parameters β_k are set to zero.

The initial spot is chosen to be $S_0 = 100$, while the actual volatility is fixed at $\sigma = 17\%$. The market-implied volatility surface used is displayed in figure 1(a) (see Lucic & Tse (2024) for the exact numerical values). Regarding the order-arrival parameters, we specify $\lambda_0^{a,n} = 252 \times 100 \times (1 - |1 - K_n/S_0|)$ for all n . If prices are quoted at the fair market level, these parameters correspond to an average of 100 market sale orders executed by the market maker per day for each at-the-money option, though this number decreases for options with extreme strikes since they are less liquid. Similarly, we assume $\kappa^{i,n} = 5 \times (1 - 5 \times |1 - K_n/S_0|)/(1\% \times \text{Vega}_0^n)$ for all n and $i \in \{b, a\}$. The value of $\kappa^{i,n}$ becomes smaller for options with extreme strikes, capturing the idea that the order flow of those options is less sensitive to a widening spread (in volatility terms). We set $\lambda_0^{b,n} = 1.15\lambda_0^{a,n}$ for all n , stipulating a scenario with net selling pressure from the market. The market-making horizon is half a trading day, such that $T = 0.5/252$. The optimal quotes are computed by (15) in conjunction with (16) and (17), where we set $\omega_n = 1\% \times \text{Vega}_0^n$ and $\gamma_n^{(k)} = 1_{\{n \in \mathcal{B}_k\}}$ for all n and k .

We conduct simulation and look at the trading profit as well as the terminal portfolio vega. We adopt time-discretisation with a step size of $\Delta t = 0.25/(60 \times 24 \times 252)$; that is, the market maker refreshes their quotes and rebalances their delta hedge every 15 seconds. To benchmark the performance of our model, we further consider three ‘zero-intelligence’ market makers of different aggressiveness who provide quotes using a constant bid-ask spread across all instruments. Three levels of constant spread are considered for each zero-intelligence market maker: 0.1 units, 0.5 units and 1 unit of percentage-vega. We use 2,000 simulation paths.

Figure 1 displays the empirical densities of the simulated terminal portfolio values and portfolio percentage-vegas. The proposed algorithmic trading rules (under different levels of the inventory penalty) outperform the zero-intelligence strategies, with higher average portfolio values (part (b)). The zero-intelligence strategy with one unit of percentage-vega spread does poorly in terms of profit and loss because the quotes are too conservative to attract enough transactions. While the zero-intelligence strategies with 0.1 and 0.5 units of percentage-vega spread achieve relatively high profit, they tend to build up a large positive and volatile vega exposure at the maturity

(as the market is a net seller of volatility under our choice of order flow parameters) – see parts (d) and (e). Under the optimal strategy, the sub-portfolios in buckets 1, 3, 4 and 6 have a relatively narrow range for the terminal vega, centred around zero, whereas those in buckets 2 and 5 have wider dispersion (part (c)). This is because we have imposed a more restrictive control over vega risk for out-of-the-money options through our choice of the baseline α_k . Modifying the values of the penalty parameters can result in different risk-return profiles attained by the proposed algorithmic market-making rule.

Concluding remarks

We introduced an option market-making model that naturally integrates the market maker’s view on the underlying volatility. Under suitable assumptions and approximations, the optimal quotes are available in a closed form that can be efficiently computed. Economically, the chosen spreads are closely tied to the volatility arbitrage profit originating from the actual-versus-implied volatility differential. We also demonstrate the model’s inherent flexibility by showcasing how market-making with multiple options written on the same underlying can be performed subject to the risk monitoring of customised factors. It is possible to further extend the current framework to incorporate additional features, such as hard position limits and market-making with general European derivatives. In terms of model limitations, one key assumption we made is that the market-implied volatility surface is unchanged throughout the market-making horizon. While this assumption is acceptable when the market-making horizon is short, consideration of more complicated implied volatility dynamics will be an interesting extension, especially if the aim is to expand the current framework to market-making with exotic derivatives. For the actual-volatility process or market order flow model, an alternative specification could be explored that can better describe empirical data. Finally, the quality of the second-order approximation technique used could also be examined more rigorously. ■

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Email: vlucic@ic.ac.uk, alex.tse@ucl.ac.uk.

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